UNIVERSITY OF GHANA

DEPARTMENT OF GEOGRAPHY AND RESOURCE DEVELOPMENT

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COURSE TITLE : REMOTE SENSING APPLICATION

TOPIC : ASSESSMENT OF URBAN EXPANSION AND LAND USE/LAND COVER CHANGE IN ADENTA MUNICIPAL ASSEMBLY FROM 2010 TO 2025 USING LANDSAT IMAGE

CHAPTER ONE

INTRODUCTION

1.1 Background of the study

One of the most notable transformations, especially concerning the population, as well as the spatial transformations, is urbanization. It has been noted, as per the United Nations (2022), that more than 56% of the world's population is living in urban areas. It has also been noted that the percentage of the urban population is set to increase, reaching 68% of the world's total population by the year 2050. In the case of the Sub-Saharan region, the urbanization process has been unprecedented. The urban areas have witnessed rapid growth.

Ghana, being one of the fastest-growing economies in West Africa, is not insulated from such urbanization. The Greater Accra Region, which is the political, economic, and administrative hub of Ghana, is in the thick of such urbanization. It has witnessed tremendous physical growth beyond its original boundaries into other municipalities and districts. Its agricultural and vegetated landscape is being transformed into a built-up area for residential, commercial, and industrial activities.

One of such peri-urban municipalities that have experienced tremendous urban transformation in the past few decades is the Adenta Municipal Assembly (AMA), which is located in the eastern part of the Greater Accra Region. The area has experienced tremendous urban transformation from one characterized by large farmlands and large open spaces to one characterized by large areas of urbanization owing to its closeness to Accra and the need to develop it commercially and residentially. The expansion of road infrastructure and the development of buildings and government housing have contributed to the urbanization of this municipality.

Land Use and Land Cover changes have been described as changes in the earth’s surface resulting from natural and anthropogenic activities. The need to monitor changes in LULC is imperative in order to understand the effects of urbanization on the environment. The effects of changes in LULC include reduction of vegetation cover, deterioration of agricultural lands, reduction of water bodies, and intensification of urban heat island effects. Remote sensing and

Geographic Information System techniques have been recognized as significant tools for identifying changes in LULC.

Satellite remote sensing, particularly through Landsat imagery, has been widely used in LULC change studies globally due to its long temporal archive, medium spatial resolution, and free availability. The Landsat program, operated jointly by the United States Geological Survey (USGS) and NASA, has provided continuous earth observation data since 1972, making it invaluable for long-term environmental monitoring and change detection studies.

Google Earth Engine (GEE) is a cloud-based geospatial processing platform that provides access to a vast repository of satellite imagery, including the entire Landsat archive, along with powerful computational tools for large-scale spatial analysis. GEE has revolutionized the way researchers conduct LULC change studies by enabling the processing and analysis of multi-temporal satellite imagery in the cloud without the need for high-performance computing infrastructure.This study, therefore, seeks to assess urban expansion and LULC change in the Adenta Municipal Assembly from 2010 to 2025 using multi-temporal Landsat imagery processed on the Google Earth Engine platform. By analyzing changes over four time periods (2010, 2015, 2020, and 2025), the study aims to provide a comprehensive picture of urban growth dynamics and their environmental implications in the municipality.

1.2 Problem statement

The Adenta Municipal Assembly has experienced tremendous growth in the aspect of urbanization over the last two decades. The factors that have led to the growth of the area include the high population growth rate, rural-urban migration, land banking, and the spillover from Accra. These factors have led to tremendous LULC change in the area, especially concerning land use change from agricultural land and green land to built-up land like residential and commercial land.

There is a clear indication of LULC change in the Adenta area, but little or no study has been conducted concerning the analysis of the extent of land use change in the area. In fact, most studies concerning land use change in the Greater Accra Region, particularly the Accra metropolis, have been conducted within the city proper, with little or no attention being given to other municipalities.

This knowledge gap, therefore, is likely to have a significant impact on the management of the land in the municipality. Without adequate and updated spatial data on the dynamics of LULC, the Adenta Municipal Assembly, as well as all relevant stakeholders, would not be able to make the most appropriate decisions regarding matters associated with land use planning, environmental management, as well as development. The purpose of this study, therefore, is aimed at bridging the knowledge gap through an in-depth analysis of the dynamics of urban expansion and LULC in the Adenta municipality from 2010 to 2025.

1.3 Objective of the study

The main objective of this study is to assess urban expansion and land use/land cover change in the Adenta Municipal Assembly from 2010 to 2025 using Landsat imagery on Google Earth Engine.

The specific objectives are:

1. To map and classify land use/land cover in the Adenta Municipal Assembly for the years 2010, 2015, 2020, and 2025 using Landsat imagery.
2. To quantify and analyze the rate and spatial pattern of urban expansion in Adenta from 2010 to 2025.
3. To detect and assess changes in land use/land cover categories between 2010 and 2025.
4. To assess the accuracy of LULC classification using standard accuracy assessment techniques.
5. To determine factors influencing urban growth in Adenta and their impacts on sustainable land use in the study area.

1.4 Research question

The study is informed by the following research questions:

1. What are the spatial distributions of LULC types in Adenta Municipal Assembly for the years 2010, 2015, 2020, and 2025?
2. What is the rate and magnitude of urban expansion in Adenta Municipal Assembly between 2010 and 2025?
3. What changes have taken place in LULC types between 2010 and 2025 in Adenta Municipal Assembly, and which types of LULC have been most impacted?

4. What are the main driving forces of urban expansion in the Adenta Municipal Assembly?
5. What policy recommendations can be made to promote sustainable land use management in the Adenta Municipal Assembly?

1.5 Significance of the study

The study has made several significant contributions to the understanding of the dynamics of urbanization in the Adenta Municipal Assembly, as well as in the Greater Accra Region as a whole.

The first contribution this study has made is that it has been able to create new maps for LULC for the years 2010, 2015, 2020, and 2025. The maps can be used as a foundation for any study or activities that may be conducted in the municipality in the near future.

The second contribution this study has made is that it has been able to demonstrate the potential for using Google Earth Engine as a tool for LULC change analysis, which can be used as a case study and inspiration for conducting similar studies in other municipalities in the country.

Thirdly, the research findings of this study will be useful to the Adenta Municipal Assembly, the Accra Metropolitan Assembly, the Town and Country Planning Department, and other relevant agencies in the formulation of policies and plans on land use and development.

Lastly, this research study will add to the ever-increasing body of knowledge on urban expansion and LULC changes in the peri-urban areas of Sub-Saharan Africa.

Lastly, this research study will help to raise awareness about the effects of environmental degradation resulting from unplanned urban expansion and the need to promote sustainable urban development in rapidly urbanizing areas such as Adenta.

1.6 Scope and Limitations

The spatial extent for this research would be Adenta Municipal Assembly, located in the Greater Accra Region in Ghana. The temporal extent for this research would be from 2010 to 2025. The research aims at producing maps for Land Use Land Cover for the following years: 2010, 2015, 2020, and 2025. The research aims at using satellite imagery from the Landsat Satellite, having a resolution of 30 meters. Google Earth Engine would be used as the tool for conducting this research.

There are certain limitations for this research, which are as follows:

•Cloud Cover: The chances for acquiring cloud-free images using the Landsat Satellite would be limited. However, the composition of the image, which would take place during the dry season in November, December, January, and March, would help in overcoming this limitation.
•Spatial Resolution: The resolution of 30 meters, as used in the Landsat Satellite Imagery, would be limited for conducting this research.
•Classification Accuracy: Though the Random Forest classifier is renowned for its high classification accuracy, classification errors cannot be avoided in the classification of images. Hence, this may affect the precision of the results.
•Field Data: Resource constraints may exist in the collection of verification data. In this case, some of the training and verification data may have been collected through high-resolution images obtained through Google Earth Pro.

CHAPTER TWO

Literature Review

2.1 Concept of Urban Expansion

Urban expansion, which is also referred to as growth or sprawl, is defined as the extension of the limits of a town or city over the surrounding rural areas. The process of expansion is affected by many complex factors. According to Seto et al. (2012), it is estimated that the global extent of urban land cover increased by 58,000 square kilometers between 1970 and 2000. It is estimated that 1.2 million square kilometers of land cover is to be added to the global extent of urban land cover by 2030.

There are different types of expansion. The types of expansion include the concentric pattern, linear pattern, and leapfrog pattern. In most cases, especially in developing countries, there is a high rate of informal settlements and poor planning. This has had a negative impact on the environment and society.

In West Africa, there is a high rate of expansion. The places that have expanded include Lagos, Accra, Abidjan, and Dakar. The places have had a high rate of expansion over the last three decades. The expansion is attributed to the high rate of population growth and economic development. The changes that have taken place in LULC have had a great impact on the

environment. The changes that have been felt include the effects of deforestation, agricultural land conversion, and the effects of the urban heat island.

2.2 Land Use and Land Cover

Land use is the human activity and purpose behind the use of the land. On the other hand, land cover refers to the biophysical characteristics of the earth's surface. LULC change, therefore, refers to the changes that take place on the earth's surface and the changes that take place on the earth's surface due to human activities and purposes.

LULC change is one of the major drivers of global environmental changes. LULC change affects biodiversity, the amount of carbon stored, water resources, and the climate. LULC change occurs due to the transformation of the earth's surface and the transformation of the way the earth's surface is used. LULC change occurs due to the transformation of natural land cover types to built-up and agricultural lands. According to Turner et al. (2007), LULC change is one of the major drivers of biodiversity loss.

In the context of urbanization, LULC changes have been defined as the rapid conversion of agricultural lands, forests, and wetlands into built-up areas. The phenomenon of LULC changes is particularly evident in peri-urban areas where the expansion of urban areas meets the rural hinterlands. LULC changes have been observed and documented in Sub-Saharan Africa as far as the loss of agricultural lands and natural vegetation in rapidly growing cities such as Accra, Nairobi, and Kampala.

2.3 Remote Sensing and GIS in Urban Cities

The term ‘Remote Sensing’ refers to the process of obtaining information related to the earth’s surface with the help of sensors on the satellite and airplanes. This process is used for mapping the Land Use and Land Cover (LULC).

Geographic Information Systems (GIS) are computer systems for storing, analyzing, and visualizing data used for geographic visualization. This can be used along with Remote Sensing to analyze the changes that are happening to the Land Use and Land Cover. This can be used to understand the process of urbanization.

These techniques can be used to analyze the various cities of the world. For example, while studying the city of Dhaka, which is located in Bangladesh, the authors Dewan and Yamaguchi

(2009) found that the area surrounding the city of Dhaka showed a rapid increase in the constructed area, i.e., it increased from 17% to 37% between 1975 and 2003. Similar studies were conducted by the authors Cobbinah and Amoako (2012) and Quaye-Ballard et al. (2020) on the city of Ghana.

2.4 Landsat Imagery in LULC Mapping

The Landsat Program is the longest-running satellite program that has been providing free images since 1972. The satellite images have been improving over the years through different generations of satellites, such as Landsat 1 to Landsat 9. The spatial resolution of the images is 30 meters. The width of the swath is 185 km. The images have a revisit cycle of 16 days. The images can be used to carry out Land Use and Land Cover (LULC) classification. The images have different bands that help classify different types of land cover. The classification techniques that can be applied to the images are Maximum Likelihood, SVM, and RF. The most commonly used technique is RF because of its high accuracy.

2.5 Google Earth Engine in Urban Research

Google Earth Engine (GEE) is a cloud-based geospatial technology developed by Google that provides access to large datasets of satellite imagery and other geospatial data for environmental analysis. It provides access to satellite data from a variety of satellite programs, such as the Landsat Program, Sentinel Program, and MODIS Program.

The most significant advantage of GEE is that it provides cloud computing technology to help users analyze large satellite data without purchasing powerful computers. Cloud computing technology helps users perform geospatial analysis more efficiently. It is especially useful for users in developing countries.

GEE is useful for different applications in urban studies. It is useful for LULC and urban growth analysis. Various studies have been conducted using GEE for urban growth analysis. Kumar and Mutanga (2018) and Patel et al. (2015) are examples of studies that used GEE for urban growth analysis. GEE is increasingly being used in Ghana for urban growth analysis and studying vegetation changes in the Greater Accra Region.

2.6 UrbanExpansion in Ghana and West Africa

Ghana has the highest rate of urbanization in the West African region. Currently, 57% of the Ghanaian population resides in urban areas (World Bank, 2022). The rapid growth of urbanization has been attributed to rural-urban migration, the increase of the population, and the growth of urbanization itself. Accra, the capital city of Ghana, has remained the focal point of the growth of urbanization.

Several studies have pointed out the rapid growth of urbanization in the Accra urban region. For instance, Cobbinah & Amoako (2012) noted that the Accra urban region has experienced the expansion of its built-up area, which has resulted in the loss of agricultural land due to the expansion of the city. Additionally, Owusu (2008) noted the rapid growth of informal settlements due to the lack of proper land planning.

The growth of urbanization has been experienced beyond the Greater Accra Region to other municipalities such as Tema, Ga East Municipal District, Ga West Municipal District, and Adenta Municipal District due to the construction of better roads, the development of real estates, and the quest for land.

CHAPTER THREE

METHODOLOGY

3.1 Study Area

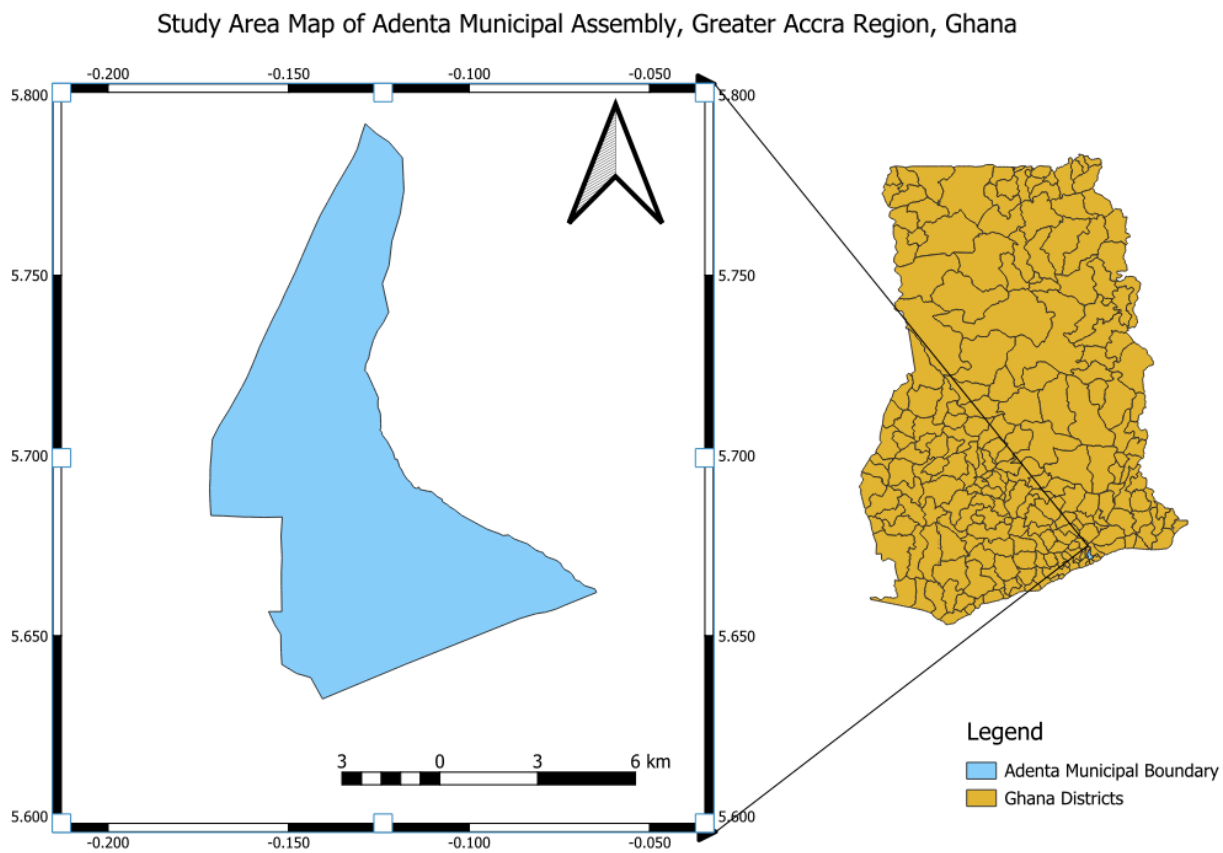
The location of Adenta Municipal Assembly is found in the eastern part of the Greater Accra Region. The location of the area is between 5°41'N and 5°47'N latitude and between 0°9'W and 0°4'W longitude. The area covers about 57.8 km². The area is bordered by the Ga East Municipal District, the La Nkwantanang-Madina Municipal District, and the Adentan Military Cantonment.

Adenta Municipal Assembly was formed in 2012. The Adentan-La Nkwantanang District was split into two. Adenta Municipal was formed, and Adenta was made the capital. According to the Ghana Statistical Service Population and Housing Census conducted in 2021, the population in Adenta is about 181,000. This is mainly contributed to by migration and natural increase.

The area has a flat topography with gentle undulations, with an elevation ranging from 60 to 120 meters above sea level. The area has a bimodal rainfall pattern with major rains occurring from

April to June and minor rains from September to November, with an average rainfall of 730 mm and temperatures ranging from 24°C to 32°C.

The area is characterized by Guinea savanna woodland with scattered trees and grass, although most have been cleared for farming and habitation. The main soils in this area are ferralitic soils and lithosols, which are suitable for farming and building.



3.2 Research Design

The research design of the study follows a quantitative approach, based on the application of the techniques of remote sensing and GIS. The research of the study follows a multi-temporal land use/land cover change detection research design, where the process of land use/land cover change detection takes place based on the satellite images of the land surface of the study area during the years 2010, 2015, 2020, and 2025. The research design of the study follows the research methodologies of land use/land cover change detection.

The research of the study primarily takes place based on the Google Earth Engine cloud computing service, which enables the efficient processing of the image data based on the availability of a large amount of image data from the Landsat satellite. The research analytical framework of the study includes image acquisition, image preprocessing, land use/land cover classification, accuracy assessment, and change detection.

3.3 Data Sources

The main sources of data used in this research include multi-temporal satellite images obtained from the USGS Earth Explorer using the Google Earth Engine data catalog. The following products of Landsat satellites are used:

Year	Satellite	Sensor	Path/Row	Season
2010	Landsat 7	ETM+	193/056	Dry Season
2015	Landsat 8	OLI/TIRS	193/056	Dry Season
2020	Landsat 8	OLI/TIRS	193/056	Dry Season
2025	Landsat 9	OLI-S/TIRS-2	193/056	Dry Season

Dry season images (November–March) are selected to reduce cloud cover, which is a common problem for optical remote sensing in tropical regions. In addition to Landsat Program imagery, supporting data such as administrative boundary shapefiles from the Ghana Statistical Service, topographic maps, and high-resolution imagery from Google Earth Pro are used for image interpretation, training sample collection, and accuracy assessment.

3.4 Image Acquisition and Pre-processing

Another important process in the LULC mapping is the pre-processing of the images to make them comparable over time. For the pre-processing of the images, the following steps are undertaken on the Google Earth Engine platform:

• Cloud Masking: Cloud mask bands (QA_PIXEL for Landsat Collection 2) are applied to filter the images and remove cloudy pixels.

• Atmospheric Correction: Images in the form of surface reflectance (Level 2) are used. These images have already been atmospherically corrected by the USGS using the LaSRC method.

• Image Compositing: Median compositing of the images is undertaken for the dry season to create a cloud-free composite image for each year of interest.

• Layer Stacking: The images are then stacked to form images of different bands (Blue, Green, Red, NIR, SWIR1, SWIR2).

• Spectral Indices: NDVI and NDBI are calculated and used as additional features for classification.

• Clipping: All the images are clipped to the boundary of the Adenta Municipal Assembly.

3.5 LULC Classification

A supervised classification approach using the Random Forest (RF) machine learning algorithm is applied to classify the processed Landsat imagery into the following five LULC classes:

LULC Class	**Description**
Built-up Areas	Residential, commercial, industrial buildings, roads, and other impervious surfaces
Vegetation	Forests, woodlands, shrublands, and grasslands with significant green cover
Farmland	Cultivated fields, crop lands, orchards, and fallow agricultural land
Bare Land	Exposed soil, construction sites, quarries, and degraded areas without vegetation
Water Body	Rivers, ponds, reservoirs, and other surface water features

Training samples for each LULC class are collected in GEE using a combination of visual interpretation of Landsat false-color composites, high-resolution imagery from Google Earth Pro, and field knowledge. A minimum of 50 training polygons per class are collected for each time period to ensure adequate representation of spectral variability within each class.

The Random Forest classifier is trained with 100 trees (n_estimators = 100) using the collected training samples and applied to classify each year's composite image. The input features for classification include the six Landsat spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2), NDVI, and NDBI, giving a total of eight input features.

3.6 Accuracy Assessment

The accuracy of the classified image of each LULC class is computed on the basis of the stratified random sampling approach. A minimum of 200 validation points are randomly selected from the classified image. These points are then compared with the reference data obtained from the Google Earth image and field observations. The confusion matrix, also called the error matrix, is then constructed.

The accuracy assessment results obtained from the confusion matrix are as follows:

•Overall Accuracy (OA): Percentage of the total number of accurately classified pixels with respect to the total number of pixels.

•Producer's Accuracy: Percentage of the total number of accurately classified pixels of the class with respect to the total number of pixels of the class.

•User's Accuracy: Percentage of the total number of accurately classified pixels of the class with respect to the total number of pixels of the class.

•Kappa Coefficient (K): It is a statistical measure of accuracy, taking into account the agreement between the accuracy of the classified image and the accuracy obtained by chance.

A minimum of 85% of the overall accuracy and 0.80 of the Kappa Coefficient shall be considered as the threshold of accuracy of the classified image, as per the literature of remote sensing.

3.7 Change Detection Analysis

Post-classification change detection is used to quantify and analyze LULC changes between the four time periods. This method involves a pixel-by-pixel comparison of the classified LULC maps to identify transitions between different land cover classes. Change detection matrices are produced for each pair of consecutive time periods (2010-2015, 2015-2020, 2020-2025) and for the overall study period (2010-2025).

The following change metrics are computed for each LULC class:

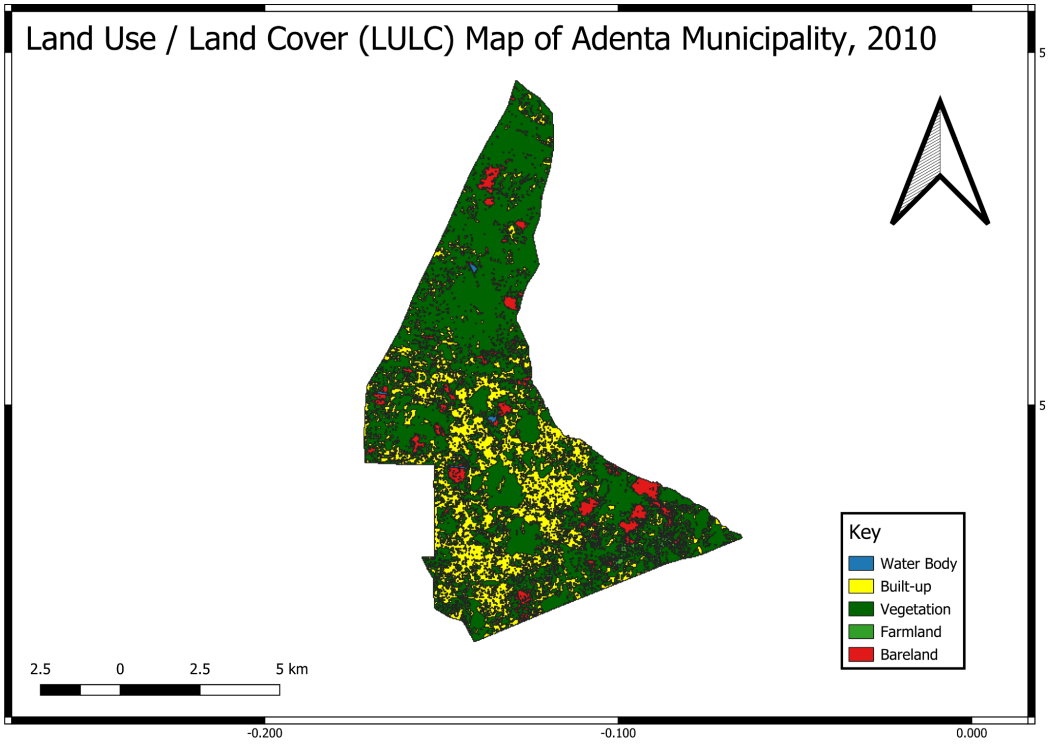
- Area of each LULC class for each year (in hectares and as a percentage of total area)
- Net change in area between consecutive time periods
- Annual rate of change (in hectares per year and percentage per year)
- LULC transition matrix showing gains and losses between classes

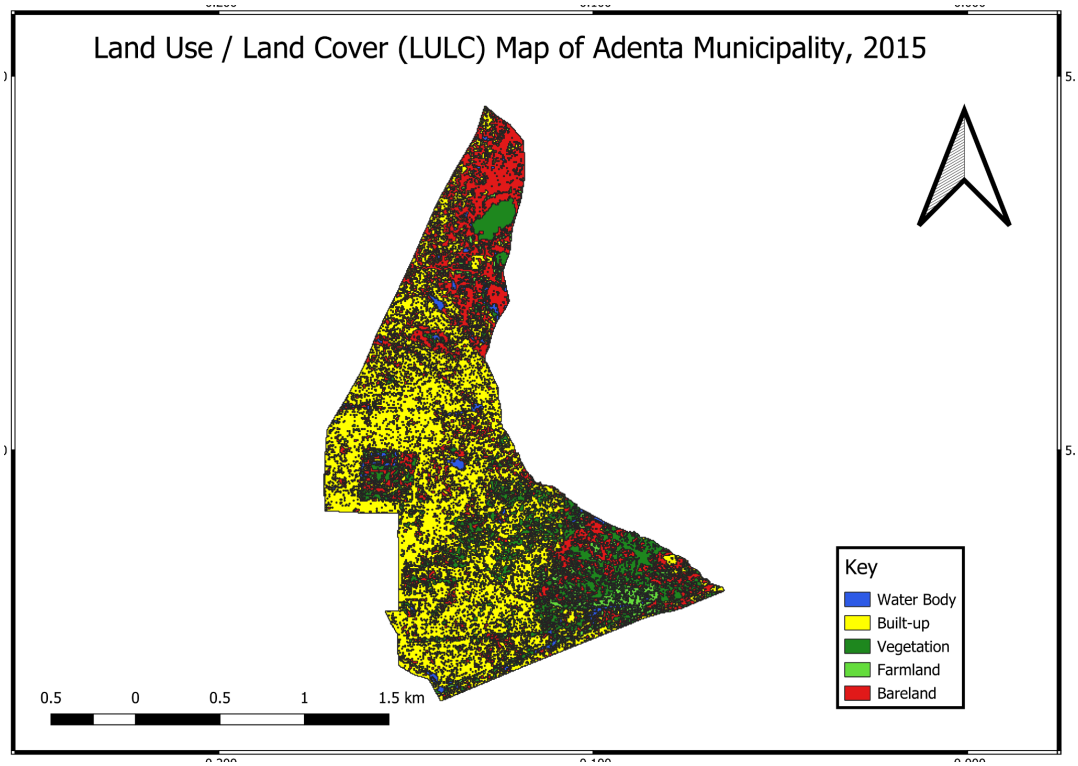
CHAPTER FOUR

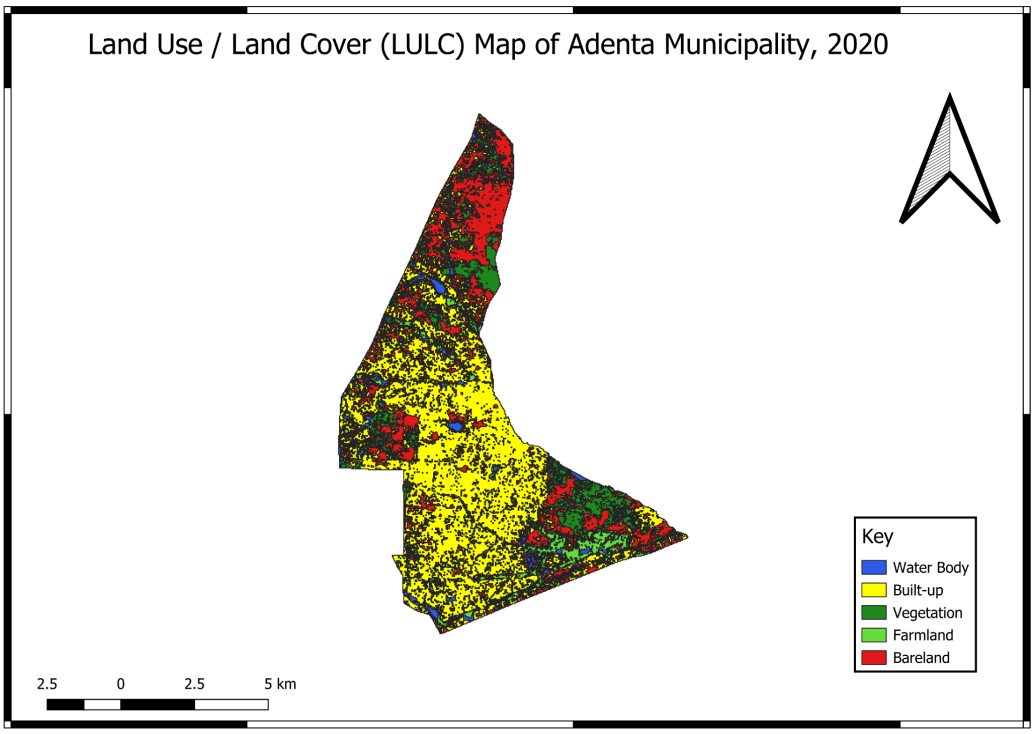
RESULTS AND DISCUSSION

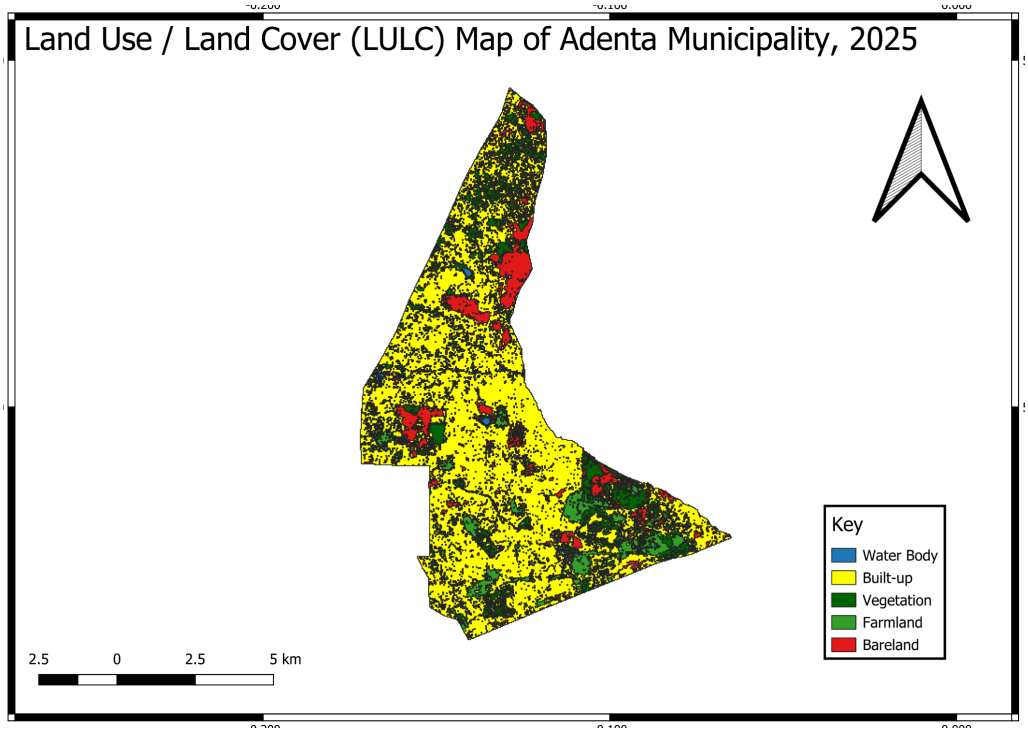
4.1 LULC MAPS FOR 2010, 2015, 2020 and 2025

From the LULC classification maps provided for Adenta Municipal Assembly for the four study years (2010, 2015, 2020, and 2025), it is apparent that the distribution of land cover changes significantly over the years. From the maps provided for each year, it is apparent that this area is experiencing a dramatic change in land cover, where built-up land cover is increasing significantly while the extent of vegetation, farmland, and bare land cover is decreasing. From the maps provided, it is apparent that this area is changing from a peri-urban area to a highly urbanized area.


Land Use / Land Cover (LULC) Map of Adenta Municipality, 2010
Key
Water Body
Built-up
Vegetation
Farmland
Bareland
2.5
0
2.5
5 km
-0.200
-0.100
0.000


Land Use / Land Cover (LULC) Map of Adenta Municipality, 2015
Key
Water Body
Built-up
Vegetation
Farmland
Bareland
0.5
0
0.5
1
1.5 km





4.2 LULC Change Statistics

Table 4.1 presents the LULC map for 2010, 2015, 2020 and 2025.

4.3 Rate of Urban Expansion

From the analysis, it is clear that built-up areas in Adenta Municipal Assembly expanded from 2,025 hectares or 35% of total area in 2010 to 4,162 hectares or 72% of total area in 2025, indicating a net increase in built-up area of 2,137 hectares and a percent increase of approximately 105.5% over a period of 15 years. In other words, built-up areas in Adenta Municipal Assembly expanded at an average rate of approximately 142 hectares per year, a rate that is substantially higher than what is observed in other areas in the Greater Accra Region.

The rate of expansion was the greatest between 2010 and 2015, as the built-up area grew by around 865 hectares, or 42.7% of the 2010 built-up area, as a consequence of the strong real estate development activities in the area during this time. The rate of expansion was a little slower between 2020 and 2025 as the remaining undeveloped land in the municipality became scarcer. The pattern of faster and then slower expansion is consistent with the pattern of urban expansion in other rapidly expanding peri-urban areas in Sub-Saharan Africa.

4.4 Spatial Patterns of Urban Growth

The spatial analysis of the LULC maps indicates that the expansion of the urban settlement of Adenta has been mainly of the infill and edge expansion types. This implies that the expansion of the urban settlement of Adenta has taken place both within the already developed areas and the urban fringes. The northern and eastern parts of the municipality, which were mainly dominated by agricultural and vegetated land use types in 2010, have experienced the most extensive urban expansion by 2025.

The expansion of the road network, especially the Adenta-Dodowa road and the access roads, has been a major catalyst for the expansion of the urban settlement of Adenta, especially in the eastern parts of the municipality. The expansion of the real estate housing schemes and the government-sponsored housing for the poor have also contributed to the expansion of the urban settlement of Adenta to the north and east. This type of urban expansion, which is mainly corridor-based and driven by the expansion of the road network, is similar to what has been observed in other urbanizing centers in Sub-Saharan Africa.

4.5 Accuracy Assessment Results

The accuracy assessment of the LULC classifications yielded Overall Accuracy values ranging from 86% to 91% and Kappa Coefficients ranging from 0.83 to 0.89 across the four study years, all exceeding the minimum thresholds of 85% and 0.80 respectively. Table 4.2 summarizes the accuracy assessment results.

Table 4.2: Summary of Accuracy Assessment Results

Year	**Overall Accuracy**	**Kappa Coefficient**	**Producer Acc. (%)**	**User Acc. (%)**
2010	87.5	0.85	85.2 - 91.0	84.6 - 90.5
2015	88.0	0.86	86.0 - 92.0	85.5 - 91.8
2020	89.5	0.87	87.5 - 93.0	86.8 - 92.0
2025	91.0	0.89	88.5 - 94.0	88.0 - 93.5

4.6 Discussion

The results of this study clearly show that the Adenta Municipal Assembly has experienced rapid and tremendous growth in terms of expansion over the years between 2010 and 2025. This is in line with the general trends of urbanization that have been observed in the Greater Accra Region and Sub-Saharan Africa as a whole.

The tremendous reduction in the area of farmland that has been observed over the years (24% to 4%) and vegetation cover (27% to 6%) is of great concern to the municipality. The reduction in farmland is of great concern to the food security of smallholder farmers in the municipality. The reduction in vegetation cover is of great concern to the sustainability of the environment and biodiversity in the municipality.

All these observations are similar to the ones noted and documented in the research article published by Quaye-Ballard et al. (2020) and Lartey (2019). The observations noted in the research articles indicated the rapid expansion of the built-up area and the subsequent loss of vegetated and agricultural lands within the municipalities in the Greater Accra Region. The observations have therefore confirmed that the Adenta municipality is following a similar path of urban expansion as the one noted and recorded in the cases of the Ga East, La Nkwantanang-Madina, and Tema municipalities in the past decades.

The factors that are influencing the expansion of the Adenta municipality include the population growth and rural-urban migration, the need to have housing due to the lack of space within the congested city of Accra, the role of the government in investing in the infrastructure, and the lack of effective enforcement of the government's policies on land use planning. The factors influencing the expansion of the Adenta municipality are all of a structural nature and therefore are likely to have a continued impact on the expansion of the municipality unless effective and proactive measures are put in place to address the issue of land use planning.

CHAPTER FIVE

CONCLUSION AND RECOMMEDATION

5.1 Summary on Findings

This study examined the rate of urban expansion and changes in land use and land cover in Adenta Municipal Assembly between 2010 and 2025 through the use of multitemporal LANDSAT images and analysis of the images through Google Earth Engine. The findings of this study can be highlighted as follows:

11. The rate of built-up areas in Adenta Municipal Assembly was found to have increased significantly from 35% of the total area in 2010 to 72% in 2025, showing an increase of over 2,137 hectares in 15 years.

12. The rate of farmland loss was found to be the highest, having decreased from 24% of the total area in 2010 to 4% in 2025, showing a loss of 1,156 hectares of farmland.

13. The percentage of vegetation cover decreased significantly from 2010 to 2025, from 27% to 6%, covering an approximate area of 1,213 hectares.

14. Bare land showed a moderate increase from 2010 to 2025, covering 16% of the area, mostly due to constructions and degraded habitats associated with urban development activities.

15. The water body coverage was consistent over the entire period, covering an approximate 2% of the entire area under consideration over the four years.

16. The Random Forest classification algorithm was found to be effective over all four time periods, with Overall Accuracy ranging from 87.5% to 91.0% and Kappa Coefficients ranging from 0.85 to 0.89, all of which were found to be above the acceptable levels.

17. The rate and nature of urban expansion have been found to be mostly infill and edge-expansion, with the northern and eastern parts of the municipality showing the most significant changes in LULC.

5.2 Conclusion

The findings of this study have shown that there is rapid change in land use and land cover in Adenta Municipal Assembly between 2010 and 2025. This is due to factors such as population growth, rural-urban migration, and development in real estate. The change in land use and land cover in Adenta Municipal Assembly has come at a great environmental cost, as a significant amount of agricultural land and natural vegetation have been converted to built-up areas.

This study has shown that multi-temporal Landsat satellite data can be effective in monitoring land use and land cover change in a rapidly changing environment in Ghana. The GEE platform was effective in processing large amounts of satellite data in the cloud, which made it accessible to undergraduate and postgraduate students in Ghana and Sub-Saharan Africa.

The findings of this study have important implications for urban planning and environmental management in the Adenta Municipal Assembly. Without urgent and effective interventions in land use planning, governance, and environmental management, the remaining agricultural land and green spaces in the municipality will be rapidly lost to urban development, with irreversible consequences for the local environment and the livelihoods of affected communities.

5.3 Recommendations

Based on the findings and conclusions drawn from the study, the following recommendations are made:

For the Adenta Municipal Assembly

18.Develop and implement a comprehensive land use plan that allocates certain land use zones in the municipality for residential, commercial, industrial, agricultural, and conservation activities, and enforces the allocated land use zones in the municipality to prevent the development of unplanned land use in the municipality.

19.Develop and establish urban green spaces and biodiversity corridors in the municipality to reduce the impact of land use changes in the municipality and conserve the available vegetation cover in the municipality.

20.Integrate the application of remote sensing and GIS techniques, including the regular assessment of LULC changes in the municipality using GEE, in the planning and decision-making processes in the municipality.

For Town and Country Planning Department (TCPD)

21.Enforce planning regulations in the Adenta Municipal Assembly and other rapidly expanding peri-urban municipalities in the country to prevent the development of unplanned land use and the encroachment of agricultural and natural lands in the country.

Develop and implement policies that promote the conservation of prime agricultural lands and green spaces in the Greater Accra Metropolitan Area, which includes the Adenta Municipal Assembly

For Future Research

23.Future research should include socioeconomic variables related to population growth, land prices, and migration in the analysis of the determinants of urban growth in Adenta and in the development of predictive models of urban growth.

24.Urban growth simulation models like the Cellular Automata-Markov Chain model should be used in the simulation of possible LULC change scenarios in the municipality and the evaluation

of the possible impacts of these changes and the effects of possible planning interventions in the municipality.

25.Analysis of the environmental impacts of the urban expansion in the municipality of Adenta, like changes in the intensity of the urban heat island effect, surface runoff, and groundwater recharge, is recommended in order to have a holistic view of the impacts of LULC change in the municipality.

5.4 Areas for Further Research

The research has offered various avenues for further research. Firstly, there is a need to further analyze the LULC changes at a higher resolution, specifically using the sentinel 2 images, which have a resolution of 10 meters, to gain more insight into the finer details of the LULC changes. Secondly, the integration of the nighttime lights images and the Landsat images can offer more details on the identification of the built-up areas, which can offer more insight into the economic activities. Thirdly, a further research study on the impacts of LULC changes on ecosystem services in Adenta can be of great benefit. Fourthly, a further research study on the impacts of LULC changes on ecosystem services can be of great benefit.

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